

Metrocar Data Analysis (SQL part)

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-- sheet 1
-- UsersFunnel

-- Downloads
SELECT count(*) FROM app_downloads;
-- Signups
SELECT count(distinct user_id) FROM signups;
-- Ride requests
SELECT count(distinct user_id) FROM ride_requests
WHERE request_ts is not NULL;
-- Ride accepted
SELECT count(distinct user_id) FROM ride_requests
WHERE accept_ts is not null;
-- Completed rides
SELECT count(distinct user_id) FROM ride_requests
WHERE dropoff_ts is not null;
-- Review left
SELECT count(distinct user_id) FROM reviews;
```

```
-- sheet 2
--Completed User Conversion

WITH
downloads AS (
    SELECT DISTINCT ad.app_download_key
    FROM app_downloads ad
),
users_who_completed_ride AS (
    SELECT DISTINCT rr.user_id
    FROM ride_requests rr
    WHERE rr.dropoff_ts IS NOT NULL
),
users_who_downloaded AS (
    SELECT DISTINCT
        s.user_id,
        d.app_download_key
    FROM downloads d
    JOIN signups s
        ON d.app_download_key = s.session_id
)
SELECT
    COUNT(DISTINCT CASE WHEN uwcr.user_id IS NOT NULL THEN uwd.user_id END) * 1.0
    / COUNT(DISTINCT uwd.user_id ) AS completed_user_conversion
FROM users_who_downloaded uwd
LEFT JOIN users_who_completed_ride uwcr ON uwd.user_id = uwcr.user_id;
```

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-- sheet 3
-- RidesFunnel
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-- Ride requests
SELECT count(distinct ride_id) FROM ride_requests
WHERE request_ts is not null;
-- Ride accepted
SELECT count(distinct ride_id) FROM ride_requests
WHERE accept_ts is not null;
-- Completed rides
SELECT count(distinct ride_id) FROM ride_requests
WHERE dropoff_ts is not null;
-- Payment
SELECT count(purchase_amount_usd)
FROM transactions
WHERE charge_status = 'Approved';
-- Review left
SELECT count(*) FROM reviews;
--
-- Cancelled rides
SELECT count(distinct ride_id) FROM ride_requests
WHERE cancel_ts is not null;

```

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-- sheet 4
--Ride Completion Rate
SELECT
    COUNT(CASE WHEN dropoff_ts IS NOT NULL THEN 1 END) * 1.0
    / COUNT(*) AS ride_completion_rate
FROM ride_requests;

```

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-- sheets 5-7
--Key Revenue KPIs
SELECT
    SUM(t.purchase_amount_usd) AS total_usd,
    SUM(t.purchase_amount_usd)/COUNT(t.ride_id) AS avg_rev_per_ride,
    SUM(t.purchase_amount_usd)/COUNT(DISTINCT rr.user_id) AS avg_rev_per_user_who_pay
FROM transactions t
JOIN ride_requests rr
    ON t.ride_id = rr.ride_id
WHERE charge_status = 'Approved';

```

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-- sheet 9
--Hour of day
EXTRACT(HOUR FROM request_ts) AS hour_of_day;

--Day of week
--(0 = Sunday, 6 = Saturday)
EXTRACT(DOW FROM request_ts) AS day_of_week;

--Ride requests by hour
SELECT
    EXTRACT(HOUR FROM request_ts) AS hour_of_day,
    COUNT(*) AS ride_requests
FROM ride_requests

```

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GROUP BY hour_of_day
ORDER BY hour_of_day;
```

```
-- sheet 10
-- hour, cancellations
SELECT
    EXTRACT(HOUR FROM request_ts) AS hour_of_day,
    COUNT(*) AS cancellations
FROM ride_requests
WHERE cancel_ts IS NOT NULL
GROUP BY hour_of_day
ORDER BY hour_of_day;
```

```
-- sheet 11
--waiting time & is_cancelled
SELECT
    ride_id,
    EXTRACT(EPOCH FROM (accept_ts - request_ts)) / 60 AS waiting_time_minutes,
    CASE
        WHEN cancel_ts IS NOT NULL THEN 1 ELSE 0
    END AS is_cancelled
FROM ride_requests
WHERE accept_ts IS NOT NULL
ORDER BY (accept_ts - request_ts) DESC;
```

```
-- sheet 12
--Downloads by platform
select
    platform,
    count(*) as downloads
from app_downloads
group by platform
order by downloads desc;
```

```
-- sheet 13
--Ride completed by platform
SELECT
    ad.platform,
    COUNT(DISTINCT rr.ride_id) AS completed_rides
FROM ride_requests rr
JOIN signups s ON rr.user_id = s.user_id
JOIN app_downloads ad ON s.session_id = ad.app_download_key
WHERE rr.dropoff_ts IS NOT NULL
GROUP BY ad.platform
ORDER BY completed_rides DESC;
```

```
-- sheet 14
-- Signups by age_range
SELECT
    age_range,
    count(*) as users
FROM signups
```

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group by age_range
order by age_range desc;
```

```
-- sheet 15
-- Ride completed users by age_range
SELECT
    age_range,
    COUNT(DISTINCT s.user_id) AS users_who_completed_ride
FROM signups s
JOIN ride_requests rr
    ON s.user_id = rr.user_id
WHERE rr.dropoff_ts IS NOT NULL
GROUP BY s.age_range
ORDER BY s.age_range;
```

```
-- sheet 16
-- Скільки поїздок у середньому робить один користувач певного віку
SELECT
    s.age_range,
    COUNT(rr.ride_id) * 1.0 / COUNT(DISTINCT s.user_id) AS avg_rides_per_user
FROM signups s
JOIN ride_requests rr
    ON s.user_id = rr.user_id
WHERE rr.dropoff_ts IS NOT NULL
GROUP BY s.age_range
ORDER BY avg_rides_per_user DESC;
```

```
-- sheet 17
-- Signups by age_range and platform
SELECT
    ad.platform,
    s.age_range,
    COUNT(DISTINCT s.user_id) as users
FROM signups s
JOIN app_downloads ad ON s.session_id = ad.app_download_key
group by ad.platform, s.age_range
ORDER BY ad.platform, s.age_range;
```

Google Spreadsheet

лінк: <https://docs.google.com/spreadsheets/d/1rrTE4PzwhUuoACjoso-PE10OtDx1YtLXiCiO0Rmofr0/edit?usp=sharing>

Power BI

лінк: <https://app.powerbi.com/view?r=eyJrIjojNDMxMzQ0MTgtMGlyZC00NDQ4LW10TAtOGI4MWlyYzNhZmVlIiwidCI6ImRmODY3OWNkLWE4MGUtNDVhOC05I>